



MSCC Operator's Guide — Remote Operation (Preliminary)

Multus SDR Control Console for Proficio, Geminus and Ultimus transceivers

Instructions based on MSCC Avalonia and MSCC NET9

This preliminary edition covers remote operation: MSCC client on one computer; radio and MSCC servers on a host on the same LAN (Windows 11 WPF client or Linux Avalonia client → Raspberry Pi or Ubuntu host). Local install, calibration and firmware are in the Linux and Windows local guides — not here.

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□ WELCOME

This guide covers **remote operation** of MSCC, the Multus SDR Control Console: you run the client on one computer, while the radio and the MSCC servers run on another computer (the **host**) on your local network. MSCC is the control program for Multus SDR, LLC transceivers: Proficio, Geminus and Ultimus, in both Legacy and MKII versions. It works only with Multus SDR equipment.

MSCC still comes in two parts:

- **The servers.** These small programs talk to the radio over USB and do the receive and transmit signal processing. They run on the **host** — the computer the radio is plugged into. Their names are `sdrcore-recv`, `sdrcore-trans` and `ms-sdr`; you will see these names in some messages.
- **The client.** This is the window you operate from. On Windows it is **MSCC for Windows** (WPF / MSCC NET9). On Linux and the Raspberry Pi it is **MSCC UI** (window title **MSCC Avalonia**).

Preferred setup: a Windows 11 client talking to Raspberry Pi (or Ubuntu) servers on the host. The same Host / Port / Remote flow also works with an Avalonia Linux client → Pi, or a Windows client → Ubuntu host.

This guide does **not** cover installing MSCC, calibrating the radio, or updating firmware on the host. Do those with the local guides first:

- *MSCC Operator's Guide — Linux and Raspberry Pi* for a Linux or Pi host (and for Avalonia as a local client).
- *MSCC Operator's Guide — Windows* for a Windows host (and for WPF as a local client).



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Windows client ready to connect remotely: Host and Port set, Auto Start / Launch Servers unchecked, Remote visible



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□ 1. Overview (READ THIS FIRST!)

Please read this section before you try a remote connection.

1. **Firmware and host first.** The radio must already be set up, calibrated and running with current firmware on the **host**. The client window title shows Core and FW after you connect. Install, calibrate and flash using the local Linux or Windows guide for that host — not this one.
2. **LAN focus for this draft.** These instructions are for a client and host on the **same local network** (same house / same Wi-Fi or Ethernet). Using remote over the public internet (port forwarding, VPN, etc.) is out of scope here. Do not expose the remote ports (especially **8888**, and the related AF ports **9100 / 9101**) directly to the open internet without a VPN or equivalent. On a typical home LAN you usually only need those ports reachable between the client and the host — see Appendix A.
3. **Servers on the host, client only on the operator PC.** The remote client **cannot** start the MSCC servers on the host. Servers must already be running on the host before you connect (**MSCC Start**, or Auto Start / Launch Servers when you are sitting at that host). On the **remote client**, leave **Auto Start / Launch / Launch Servers unchecked** — you are only connecting to servers that are already running elsewhere.
4. **Default remote port is 8888.** Remote AF also uses **9100** (phones/digital playback toward the client) and **9101** (microphone transmit to the host). See Appendix A.
5. **Version numbers in this guide are examples.** Names such as `mscc-ui_0.6.xx`, `mscc-net9-R9-26-6`, or a title bar reading MSCC Avalonia 0.6.67 / MSCC WPF 9.27.0 stand for whatever version you were given. Always use the newest files you have.
Screenshots are examples only — the Avalonia or WPF chrome (title bar version) in a picture may not match the release you are running.
6. **Bug reports and feature requests:** post them on our forum, <https://multusdr.groups.io/g/main>



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□ 2. What you need

- A **host** computer with the radio plugged in, MSCC installed and set up (MSCC Init on Linux/Pi; local Windows setup on a Windows host), and the **MSCC servers running**.
- A **client** PC on the same LAN (or a VPN into that LAN):
 - **Windows 11** with MSCC for Windows (WPF / NET9), or
 - **Linux** with MSCC Avalonia (MSCC UI).
- The host's **IP address** and port **8888**.
- For **Digital** modes (WSJT-X etc.):
 - On Windows: a virtual audio cable such as **VB-Audio Virtual Cable** (VAC), with CABLE Input / CABLE Output.
 - On Linux: MSCC's virtual sinks (**VirtualA** / related devices from MSCC Init).
- **WSJT-X** (optional) on the client for FT8 and similar modes.

Your IP addresses will differ. The pictures in this guide use example LAN addresses from one test network (Ubuntu / stew-HP host **192.168.1.234**, Raspberry Pi host **192.168.1.193**). Use the addresses your host reports — not these examples.

□ 3. On the host

Do this on the computer the radio is plugged into.

3.1 Start the MSCC servers

1. Power the radio on and confirm its USB cable is connected to the host.
2. Start the servers:
 - **Linux / Raspberry Pi:** use **MSCC Start** from the MSCC menu (or the equivalent you use when operating locally).
 - **Windows host:** start MSCC locally with **Launch Servers / Auto Start** as in the Windows local guide, or otherwise leave the servers running before you walk over to the client.
3. Leave the servers running. You do not need the client open on the host while you operate from elsewhere.
4. **Auto Start on the host is local only.** If you turn on Auto Start / Launch Servers on the **host**, that only helps those servers come up after a reboot **on that machine**. It is not a remote-start from the client — the client still cannot start host servers over the network.

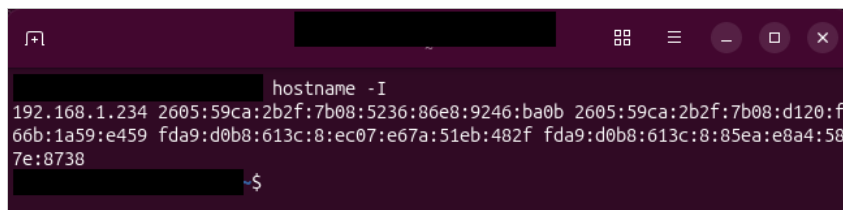
3.2 Find the host IP address

The client needs the host's LAN IP.

Linux PC or Raspberry Pi — open a Terminal and run:

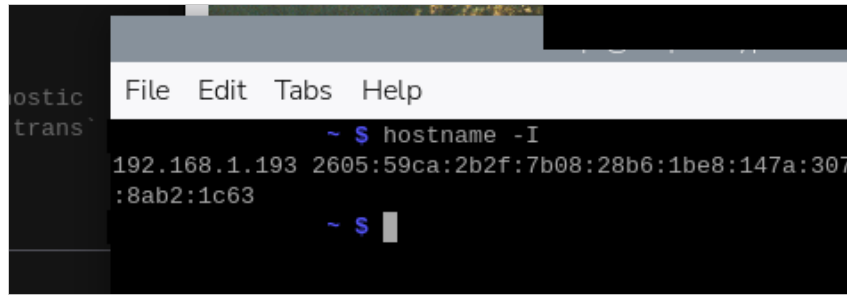
```
hostname -I
```

The first address is usually the LAN IPv4 address (for example 192.168.1.234 on an Ubuntu host, or 192.168.1.193 on a Pi). Ignore the long IPv6 addresses for this guide unless you know you need them.

A screenshot of a terminal window with a dark purple background. The terminal shows the command 'hostname -I' being executed. The output displays the IPv4 address '192.168.1.234' followed by two long IPv6 addresses. The first line of output is '192.168.1.234 2605:59ca:2b2f:7b08:5236:86e8:9246:ba0b 2605:59ca:2b2f:7b08:d120:f66b:1a59:e459 fda9:d0b8:613c:8:ec07:e67a:51eb:482f fda9:d0b8:613c:8:85ea:e8a4:587e:8738'. The prompt is '-\$'.

Ubuntu host: hostname -I shows the LAN address (example 192.168.1.234). Login name and hostname are blacked out.

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A screenshot of a terminal window on a Raspberry Pi. The window has a menu bar with 'File', 'Edit', 'Tabs', and 'Help'. The terminal shows a prompt '~ \$' followed by the command 'hostname -I'. The output is '192.168.1.193 2605:59ca:2b2f:7b08:28b6:1be8:147a:307:8ab2:1c63'. The prompt '~ \$' is followed by a cursor.

Raspberry Pi: hostname -I in a Terminal (example 192.168.1.193). Cropped to the Terminal only.

Windows host — open **Settings** → **Network & internet** and note the IPv4 address for Wi-Fi or Ethernet, or run `ipconfig` in Command Prompt / PowerShell and read the IPv4 Address for that adapter.

Write the address down; you will type it into the client's **Host** field.



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☐ 4. On the Windows client (WPF / MSCC NET9)

These steps use MSCC for Windows on a PC that is **not** the radio host. Instructions are based on MSCC Avalonia and MSCC NET9; the WPF pictures here show an example build such as MSCC 9.27.0 (title-bar version in shots may not match your release).

4.1 Prepare the client

1. Install MSCC for Windows on the client PC if it is not already there (Windows local guide). You do **not** need the radio plugged into the client.
2. For Digital modes, install **VB-Audio Virtual Cable** (or your chosen VAC) so **CABLE Input** and **CABLE Output** appear in Windows sound devices.
3. Start **MSCC** on the client.

4.2 Host, Port, and Start

1. On the left **SERVER** panel, **uncheck Auto Start** and **uncheck Launch Servers**. The servers must run on the host, not on this PC.
2. Set **Host** to the host's LAN IP (example 192.168.1.234).
3. Set **Port** to **8888** (default).
4. Press **Start**.

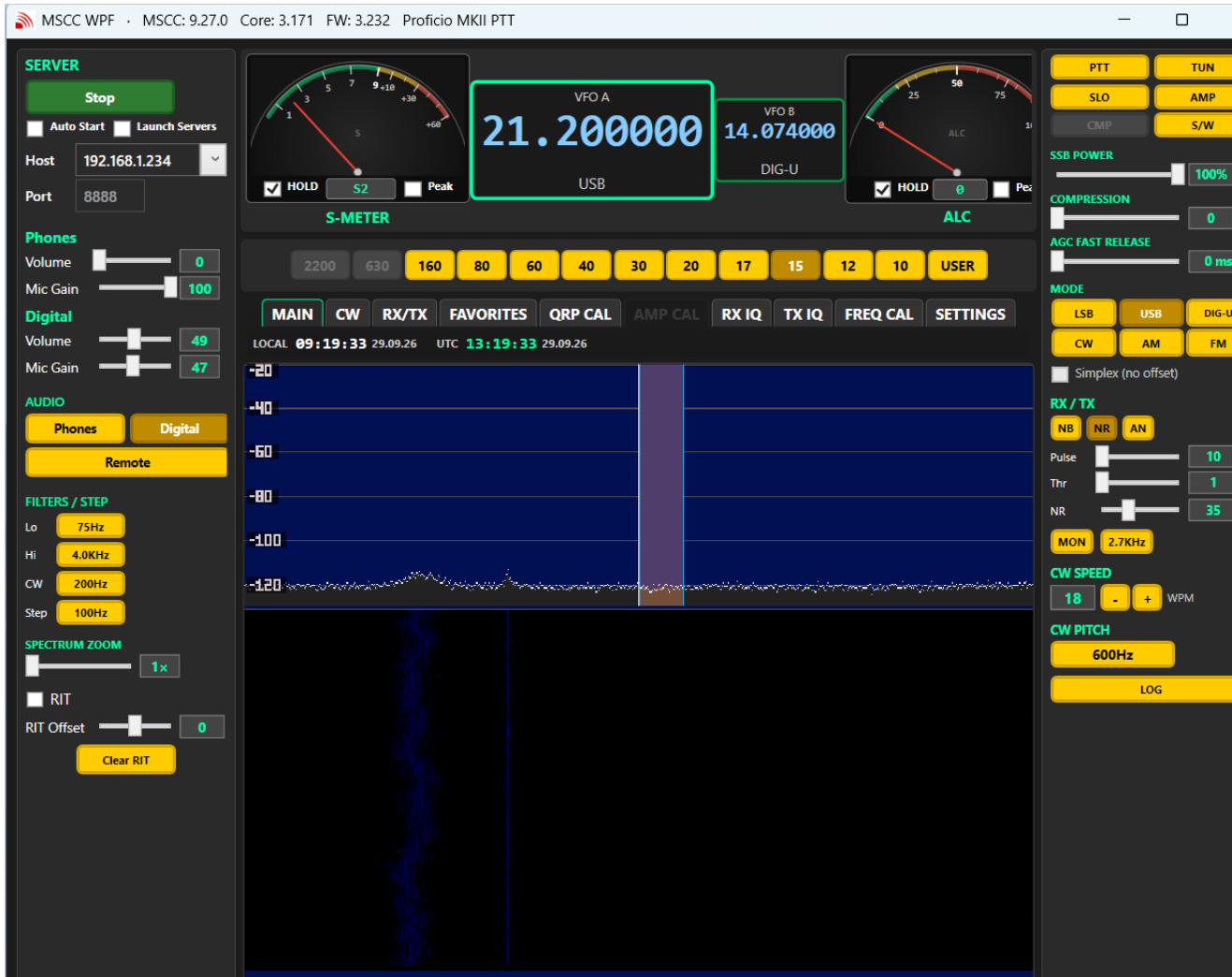
When the connection works, the button becomes **Stop**, the spectrum and waterfall come alive, and the title bar shows Core and FW from the host radio.

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Before connect: Host and Port entered, Auto Start / Launch Servers unchecked, VFO blank, Remote button visible

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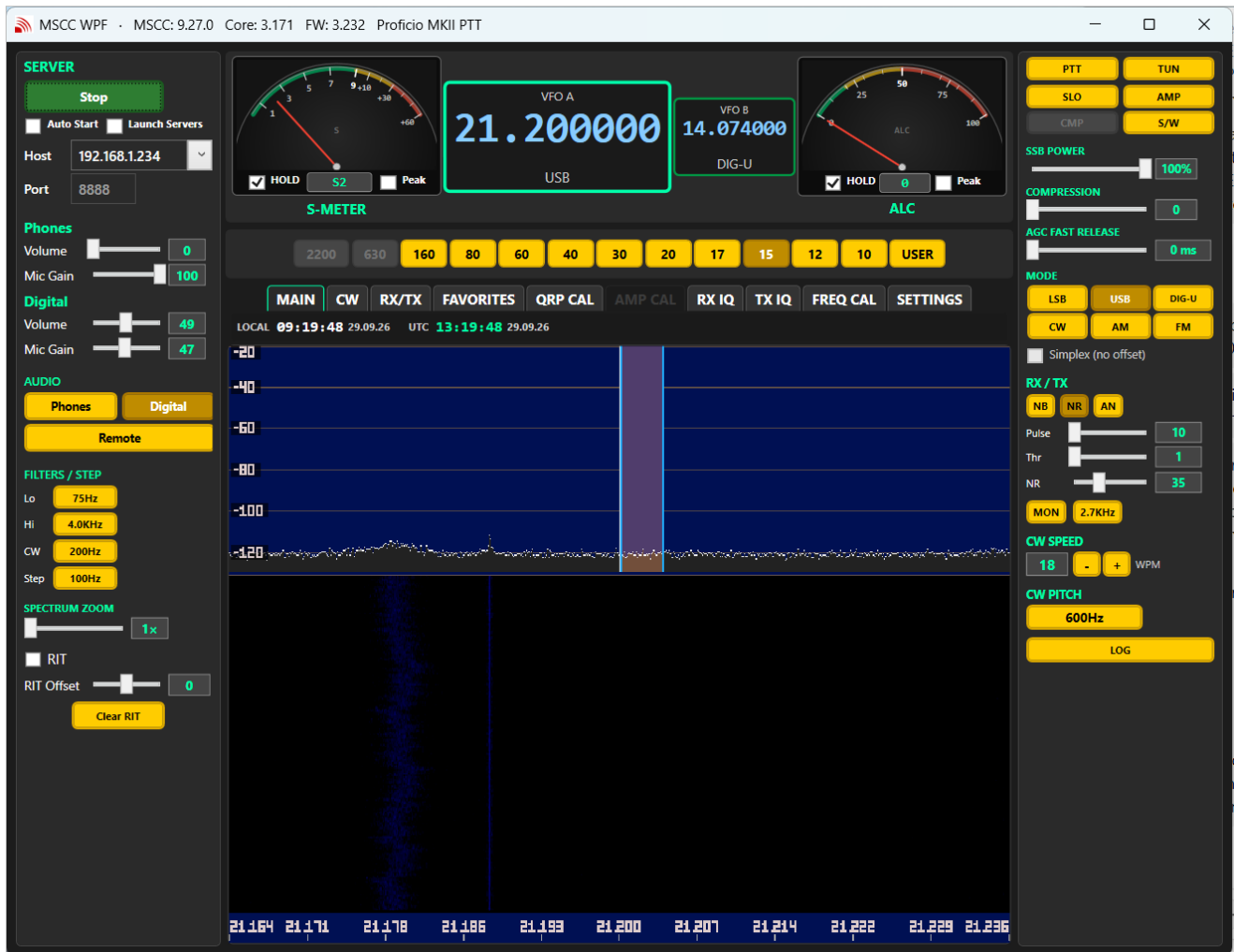


Connected with Remote and Phones on: live spectrum, Core/FW in the title bar

4.3 Remote audio: Phones vs Digital

1. Press **Remote** on the left **AUDIO** section. Remote is for operating when the radio is on another computer. It opens the Remote Audio window (Remote Phones or Remote Digital).
2. Choose the path you need:
 - **Phones** — hear the radio on this PC's speakers or headset; use this PC's microphone for voice TX.
 - **Digital** — route audio through VAC for WSJT-X (or similar). You can light **Phones** and **Digital** together when you need both paths.

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Connected with Remote, Phones and Digital lit

Remote stays available only for a remote Host. If Host is 127.0.0.1 or Launch Servers is checked, Remote is for local use rules in the Windows guide (often grayed). For remote, keep Launch Servers off and Host set to the other computer's IP.

4.4 Remote Phones window

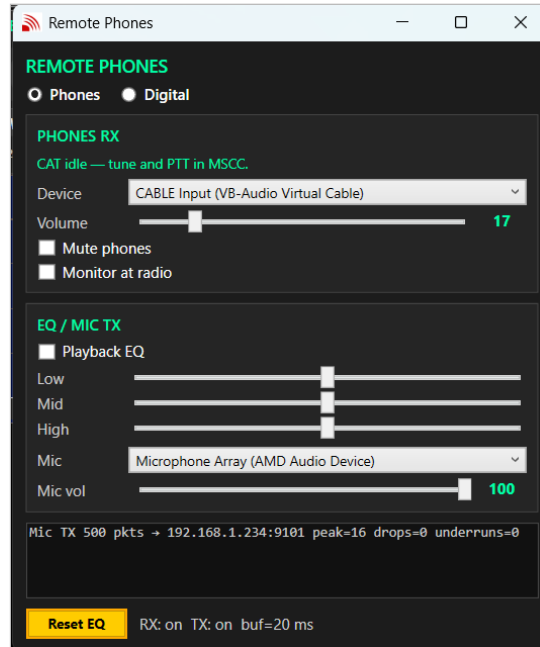
With **Remote** and **Phones** selected, the **Remote Phones** window sets playback and mic on this PC.

Typical voice / phones setup:

- **Device** (playback): your headset or speakers — or CABLE Input if you are feeding another program. The example shot uses **CABLE Input (VB-Audio Virtual Cable)** while testing.

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- **Mic**: this PC's microphone (example: Microphone Array).
- Status lines show **Mic TX ... → :9101** when microphone audio is being sent to the host. Port **9101** is the remote AF mic TX port.



Remote Phones window: playback device, mic, Mic TX to host:9101

4.5 Remote Digital window and VAC wiring

With **Remote** and **Digital** selected, the **Remote Digital** window is for WSJT-X (and similar) on this PC.

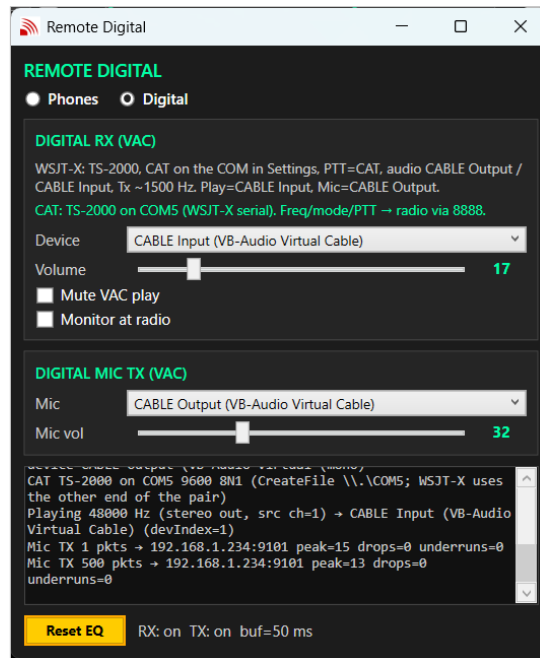
Typical VAC wiring on Windows:

Path	Device
MSCC Digital playback (radio → WSJT-X)	CABLE Input (VB-Audio Virtual Cable)
MSCC Digital mic (WSJT-X → radio)	CABLE Output (VB-Audio Virtual Cable)
WSJT-X Input	CABLE Output
WSJT-X Output	CABLE Input

The Remote Digital hints also remind you: WSJT-X as **Kenwood TS-2000**, CAT on the COM port shown in Settings, **PTT = CAT**, audio CABLE Output / CABLE Input, Tx tone near 1500 Hz. CAT/freq/mode/PTT go to the radio through the MSCC connection on port **8888**; mic audio for remote AF uses **:9101**.

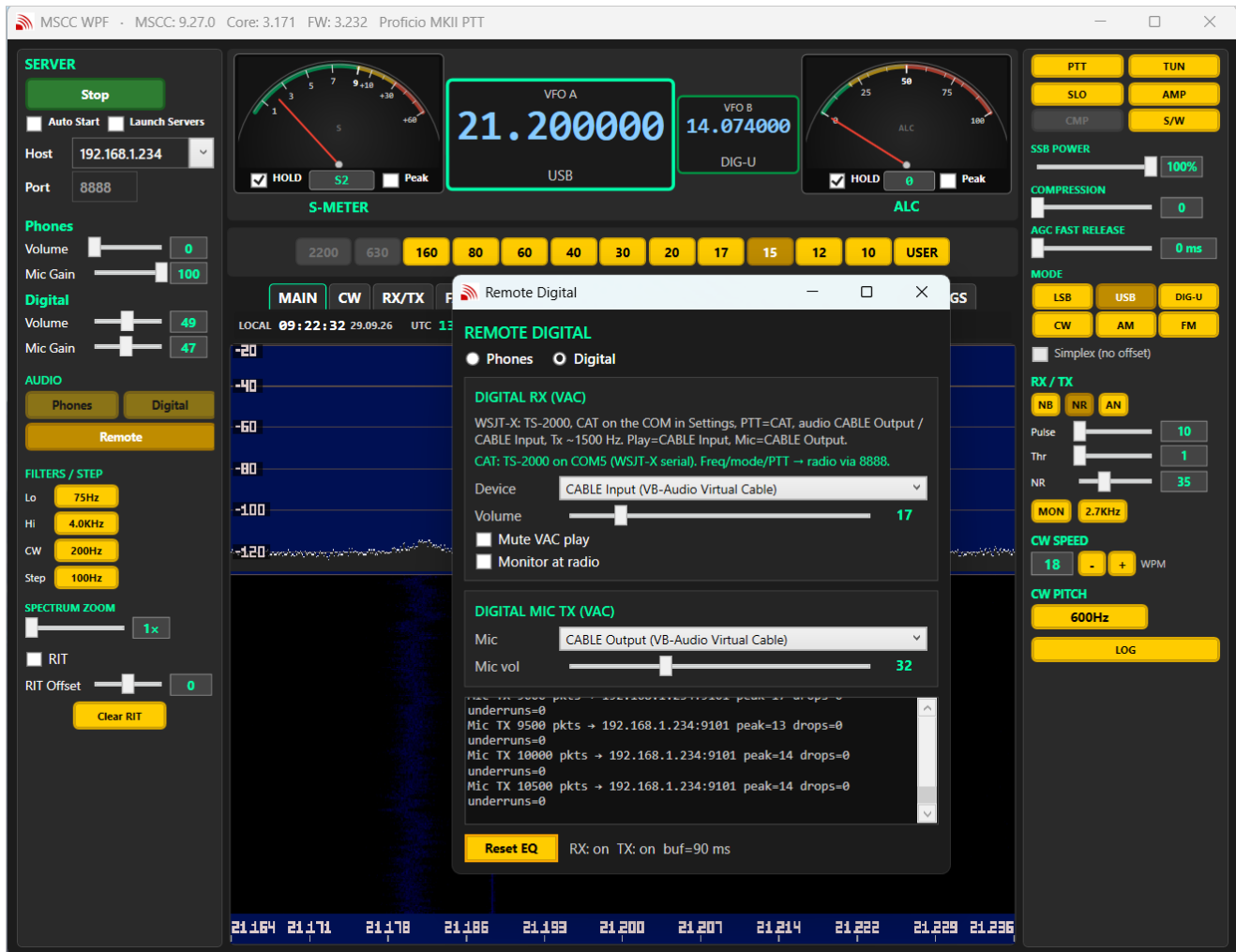


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Remote Digital window: VAC devices, CAT COM hint, Mic TX to :9101

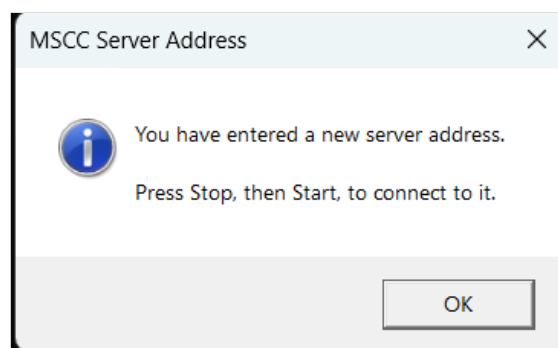
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Remote Digital open over the main WPF window while connected

4.6 Changing Host while connected

If you edit **Host** (or Port) while the client is already started, MSCC shows a dialog: you have entered a new server address — press **Stop**, then **Start**, to connect to it.



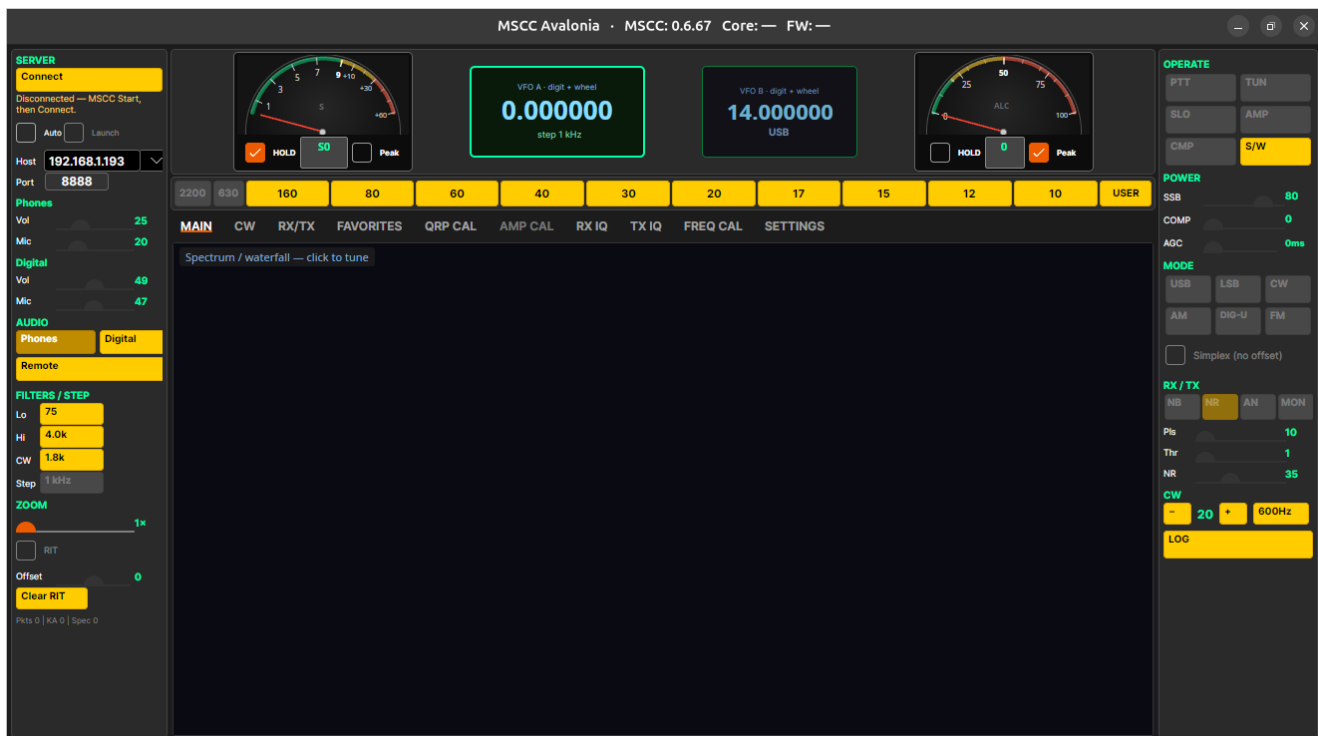
Dialog: new server address — press Stop, then Start

□ 5. On the Linux / Avalonia client

The Host / Port / Remote flow on MSCC Avalonia matches the Windows client. These pictures use an Ubuntu client connecting to a Pi host at example **192.168.1.193**. Title-bar versions in the Avalonia shots (for example 0.6.67) are examples only and may not match your current release.

5.1 Connect

1. On the host (Pi or Ubuntu), servers must already be running (**MSCC Start**).
2. On the client, start **MSCC UI**.
3. **Uncheck Auto** and **uncheck Launch**.
4. Set **Host** to the host IP (example 192.168.1.193) and **Port** to **8888**.
5. Press **Connect**. Status should read **Connected** — **spectrum OK**, and Core / FW appear in the title bar.
6. Press **Remote**, and use **Phones** and/or **Digital** as on Windows.



Avalonia pre-connect to Pi: Host set, Auto/Launch unchecked, Remote lit

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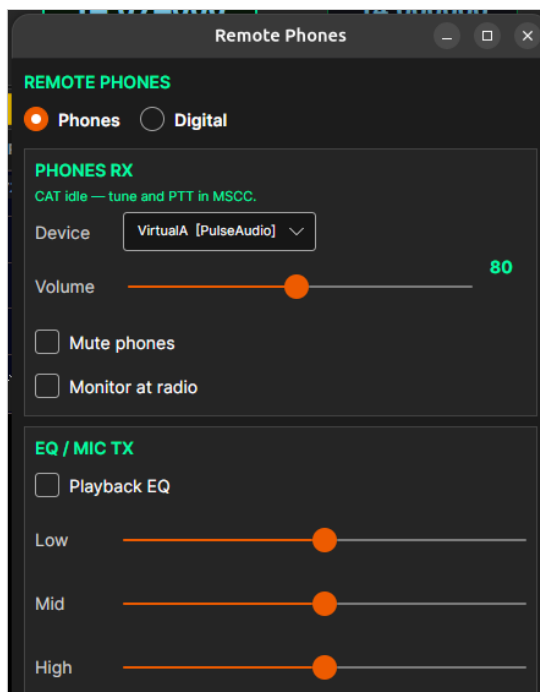


Avalonia connected with Remote + Phones, DIG-U, live spectrum

5.2 Remote Phones on Avalonia (VirtualA)

On Linux, operator / remote phones playback often uses MSCC's **VirtualA** PulseAudio device (created with the MSCC install / Init), rather than VB-Audio.

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Avalonia Remote Phones window with Device VirtualA

5.3 Suggested COM / tnt pairings (Avalonia only)

On a **Linux or Pi** install, MSCC Init sets up /dev/tnt pairs for Kenwood CAT. Suggested COM/tnt pairings for WSJT-X in the Linux local guide apply when WSJT-X runs **on that same Linux/Pi machine**.

Avalonia-only note. Those /dev/tnt0 ↔ /dev/tnt1 pairings are for the Linux/Avalonia (and Pi) stack. They are **not** the Windows COM-pair story. A **Windows client → Pi host** remote session still uses Host / Port **8888** and Remote the same way; WSJT-X on the **Windows** client uses Windows COM ports / VAC as in Section 4 and Section 6 — not the Pi's /dev/tnt devices.

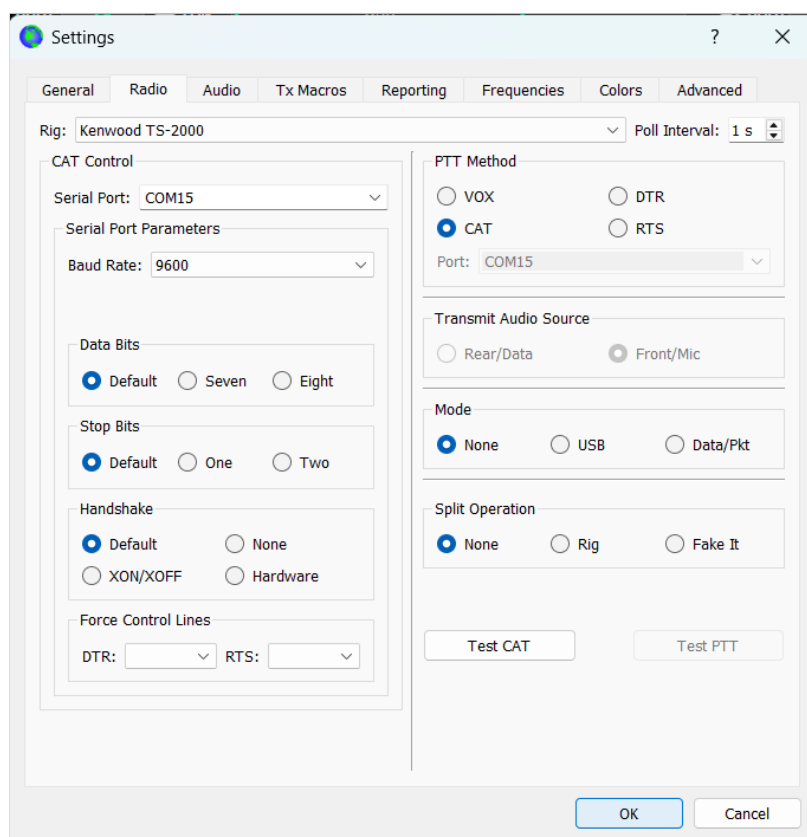
Question for Stew: for Win client → Pi host digital, does WSJT-X CAT always stay on the Windows COM pair that MSCC Remote Digital opens, with the Pi only seeing CAT via the 8888 path — confirm wording.

□ 6. Digi with WSJT-X (remote client)

These steps assume the client is already connected with **Remote** and **Digital** on, VAC (Windows) or virtual sinks (Linux) wired, and the host servers running.

6.1 Mode in MSCC and in WSJT-X

- In **MSCC**, select **DIG-U** (or **DIG-CW** when that is what you use).
- In **WSJT-X** → **Settings** → **Radio**, Stew's preferred setup is **Mode: None**, so **MSCC owns the mode**. Do not let WSJT-X fight MSCC over USB/LSB/DIG.
- **Rig:** Kenwood TS-2000.
- **PTT Method:** CAT.
- **Serial Port:** the COM port that pairs with MSCC's CAT (example shots show COM15 on the Radio tab; Remote Digital hints may mention another COM — use the pair MSCC opens on your PC). **To confirm:** document the exact COM pair MSCC NET9 creates for Remote Digital on Windows.

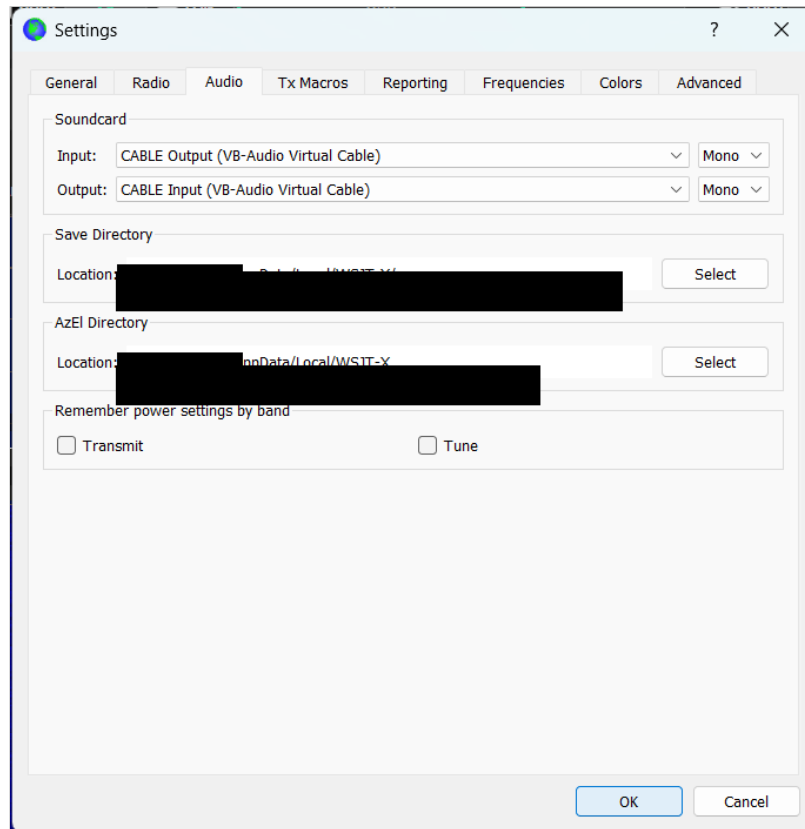


WSJT-X Radio tab: TS-2000, CAT, Mode None, Split None

6.2 Audio in WSJT-X

On Windows with VB-Audio:

- **Input:** CABLE Output (VB-Audio Virtual Cable)
- **Output:** CABLE Input (VB-Audio Virtual Cable)

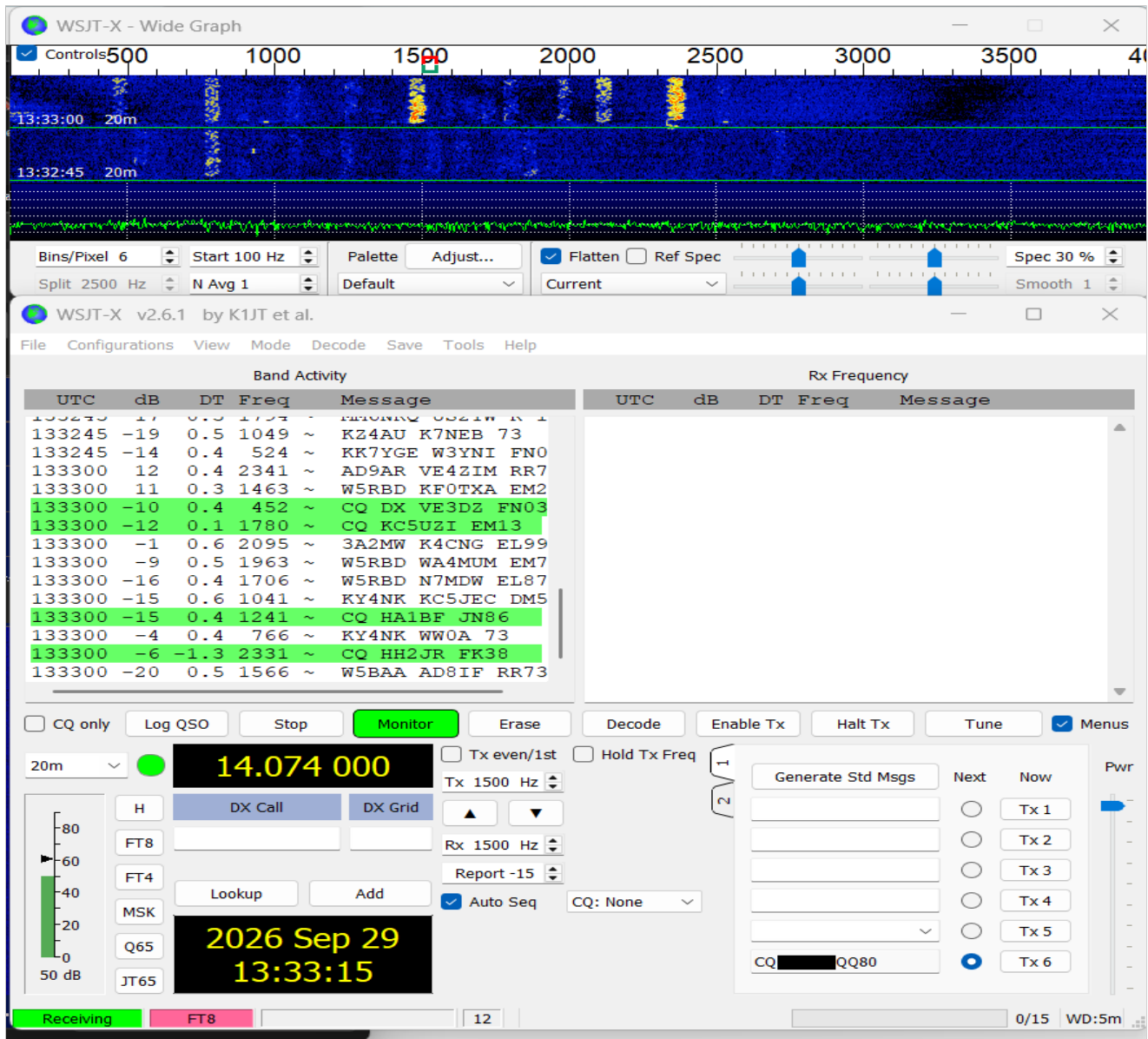


WSJT-X Audio tab: CABLE Output in, CABLE Input out

6.3 Operate

1. In MSCC, tune to the digi frequency (for example 14.074 on 20 m) and set **DIG-U**.
2. In WSJT-X, select the band / frequency and **Monitor**.
3. Decodes should appear when the path is correct. Transmit only when you are allowed to on that frequency and with ALC under control (use Digital Mic vol in Remote Digital if ALC is high).

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WSJT-X main window decoding on 20 m FT8 while MSCC runs remotely



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□ 7. Troubleshooting

Symptom	Things to check
Client will not connect	Are servers already running on the host (MSCC Start)? The client cannot remote-start them. Is Host the correct LAN IP? Is Port 8888 ? Same LAN (or VPN)? Firewall on host or client blocking UDP 8888 ?
Connects but no audio	Is Remote on? Did you pick Phones vs Digital for what you need? Is the Remote Audio window still open? Correct playback / mic / VAC device?
Digital / WSJT-X silent	VAC direction (Input vs Output) swapped? WSJT-X Audio devices match Section 6.2? Remote Digital Mic TX line showing packets to <i>host:9101</i> ?
Host field changed but still on old server	Press Stop , then Start (see the “new server address” dialog).
Remote button grayed / unavailable	Host is 127.0.0.1, or Launch Servers / Launch is checked — clear those for remote.
Spectrum OK but Mic TX fails	Host firewall allowing UDP 9101 inbound (remote AF mic TX)? Client firewall allowing UDP 9100 inbound (remote phones/digital playback from the host)?



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□ Appendix A: Port summary

Remote operation uses these **UDP** ports (from the MSCC host/server and client remote-audio code). CAT / freq / mode / PTT for a remote session ride on **8888** with the control path — there is no separate remote CAT TCP port for the operator to open.

Port	Direction (typical)	Use
8888 UDP	Client ↔ host (host listens)	Default MSCC control / spectrum / CAT path (Host field + Port).
9100 UDP	Host → client (client listens)	Remote AF playback (phones / digital RX) from sdrcore-recv to the client Remote Audio window.
9101 UDP	Client → host (host listens)	Remote AF microphone transmit (Mic TX from Remote Phones / Remote Digital to sdrcore-trans on the host).

On the **host**, operators commonly need **8888** and **9101** reachable from the client (and related replies). On the **client**, **9100** must be reachable from the host for remote phones/digital playback. Keep all of these on the LAN (or inside a VPN) — do not expose them directly to the open internet.

No other remote operator ports turned up in the station code for this path. (Internal host-only ports between the MSCC server programs, such as local **9000**, are not something the remote operator configures.)



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☐ **Appendix B: Related guides**

- Local install, MSCC Init, calibration, and firmware: *MSCC Operator's Guide — Linux and Raspberry Pi* and *MSCC Operator's Guide — Windows*.
- This remote guide assumes those steps are already done on the host.